4 Using Date and Time Form Helpers

You can choose not to use the form helpers generating HTML5 date and time input fields and use the alternative date and time helpers. These date and time helpers differ from all the other form helpers in two important respects:

- Dates and times are not representable by a single input element. Instead you have several, one for each component (year, month, day etc.) and so there is no single value in your `params` hash with your date or time.
- Other helpers use the `_tag` suffix to indicate whether a helper is a barebones helper or one that operates on model objects. With dates and times, `select_date`, `select_time` and `select_datetime` are the barebones helpers, `date_select`, `time_select` and `datetime_select` are the equivalent model object helpers.

Both of these families of helpers will create a series of select boxes for the different components (year, month, day etc.).

4.1 Barebones Helpers

The `select_*` family of helpers take as their first argument an instance of `Date`, `Time` or `DateTime`that is used as the currently selected value. You may omit this parameter, in which case the current date is used. For example:

```
<%= select_date Date.today, prefix: :start_date %>
```

outputs (with actual option values omitted for brevity)

```
<select id="start_date_year" name="start_date[year]"> ... </select>
<select id="start_date_month" name="start_date[month]"> ... </select>
<select id="start_date_day" name="start_date[day]"> ... </select>
```

The above inputs would result in `params[:start_date]` being a hash with keys `:year`, `:month`, `:day`. To get an actual `Date`, `Time` or `DateTime` object you would have to extract these values and pass them to the appropriate constructor, for example:

```
Date.civil(params[:start_date][:year].to_i,
params[:start_date][:month].to_i, params[:start_date][:day].to_i)
```

The `:prefix` option is the key used to retrieve the hash of date components from the `params` hash. Here it was set to `start_date`, if omitted it will default to `date`.

4.2 Model Object Helpers

`select_date` does not work well with forms that update or create Active Record objects as Active Record expects each element of the `params` hash to correspond to one attribute. The model object helpers for dates and times submit parameters with special names; when Active Record sees parameters with such names it knows they must be combined with the other parameters and given to a constructor appropriate to the column type. For example:

```
<%= date_select :person, :birth_date %>
```

outputs (with actual option values omitted for brevity)

```
<select id="person_birth_date_1i" name="person[birth_date(1i)]"> ...
</select>
<select id="person_birth_date_2i" name="person[birth_date(2i)]"> ...
</select>
<select id="person_birth_date_3i" name="person[birth_date(3i)]"> ...
</select>
```